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Summary of Savannah River Site FY24 Salt Waste Qualification Data

K. P. Crapse

T. B. Peters

October 2025

SRNL-STI-2025-00489, Revision 0

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Printed in the United States of America

Prepared for
U.S. Department of Energy

SRNL-STI-2025-00489
Revision 0

Keywords: *SWPF, Salt Waste Processing Facility*

Retention: *Permanent*

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Savannah River National Laboratory is operated by Battelle Savannah River Alliance for the U.S. Department of Energy under Contract No. 89303321CEM000080.

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EXECUTIVE SUMMARY

Savannah River Mission Completion (SRMC), the Liquid Waste Operations subcontractor at SRS, and Savannah River National Laboratory (SRNL) analyzed samples from Savannah River Site (SRS) Waste Tanks 21H, 41H, and 42H to support qualification of Salt Waste Processing Facility (SWPF) Waste Batches 10, 11, 12, and 13 for processing (the FY24 Salt Batch Qualification samples). These Tanks (i.e. 21H, 41H and 42H) are blend tanks for feed to SWPF. This report focuses on the characterization of the Salt Batch Qualification sub-samples (i.e., any adjustments after qualification or as a part of processing are documented elsewhere). None of the samples displayed any unusual or unexpected characteristics such as large amounts of solids, floating solids, or unusual color. Characterization of these samples confirmed similar chemical composition and characteristics to previous salt waste batches. The analytical results (both rapid, typically 4 weeks, and long-term, typically 8 weeks) for Salt Batches 10, 11, 12, and 13 are now summarized and discussed in this technical report. SRMC-Analytical Laboratory (SRMC-AL), for the first time, provided all the short-term analyses data used in the qualification evaluations of SWPF Salt Batches 10, 11, 12, and 13. These SRMC short-term results are presented along with the results for the long-term analyses provided by SRNL.

SWPF Salt Batch 10, 11, 12, and 13 Qualification samples were similar in chemical composition to each other and to the previous 19 Salt Waste Batches (10 Interim Salt Waste Processing (ISWP) Macrobatch and 9 SWPF Salt Batch Qualification samples) analyzed to date with some results trending at or beyond the higher end of the ranges previously observed for some analytes. Salt Batch 10, 11, 12 and 13 densities remained very close to 1.3 g/mL and the values measured for insoluble solids remained less than 200 mg/L for all batches except Batch 11 which was 550 mg/L and contained visible solids which settled with time. Salt Batch 10, 11, 12 and 13 results for sodium (6.82 to 8.26 M), aluminum (0.274 to 0.356 M), and hydroxide (3.12 to 3.85 M) continue to trend toward the high end or higher than results for salt batches analyzed to date (previous maximum to date 7.50 M (Batch 6) for sodium, 0.437 M (Batch 8) for aluminum, and 4.07 M (Batch 8) for hydroxide. Sodium and fluoride in Batch 12 (8.26 M and 0.00616 M, respectively) were measured at values higher than the previous maximum values for those analytes reported for Salt Batch Qualification samples. Total mercury values (38.0 to 50.0 mg/L) were in the middle of the range previously reported (9.17 to 129.0 mg/L). Sulfate values for Salt Batches 11 and 12 (0.0233 M and 0.0316 M, respectively) were lower than the previous range (0.05 M to 0.16 M) for all previous Salt Waste Batches. Cs-137 activities in filtered samples of Salt Batches 10, 11, 12 and 13 ranged from 3.65E+08 pCi/mL (8.1E+08 dpm/mL; 1.4 Ci/gal) to 5.73E+08 pCi/mL (1.3E+09 dpm/mL; 2.2 Ci/gal). The previous maximum value was for unfiltered Salt Batch 8 (6.22E+08 pCi/mL (1.4E+09 dpm/mL; 2.4 Ci/gal)). These values remain below the current WAC value of 3.0 Ci/gal. The total beta activity was similar to the Cs-137 values measured. In comparison of filtered and unfiltered results where available for the various radionuclide measurements, typically the filtered samples values measured were the same or lower than the unfiltered. For Batch 11 there were larger differences between filtered and unfiltered sample results likely owing to the presence of solids with which the radionuclides may be associated. I-129 (<45.5 pCi/mL to 155 pCi/mL) and Tc-99 (9.38 E+04 to 1.52E+05) values continue to trend near the WAC limits for the Saltstone Production Facility (SPF). Although the concentrations and activities continue to trend on the high end or higher than the SWPF WAC for sodium, aluminum, hydroxide, and Cs-137 (and related analyses for total gamma) as well as the downstream WAC limit for I-129 and Tc-99, no significant impact was deduced from the results based on evaluations carried out to justify any WAC deviations.

TABLE OF CONTENTS

LIST OF TABLES vii
LIST OF FIGURES vii
LIST OF ABBREVIATIONS viii
1.0 Introduction 1
2.0 Experimental Procedure 1
2.1 Sample Receipt and Preparation 1
2.2 Quality Assurance 3
3.0 Characterization Results for the FY24 SWPF Salt Batches (10, 11, 12 and 13) including analyses by SRNL and SWPF-AL 3
4.0 Comparison with Previous Salt Batch Characterization Results 7
5.0 Conclusions 10
6.0 Recommendations, Path Forward or Future Work 11
7.0 References 12

LIST OF TABLES

Table 2-1. Salt Batch Sample Description and Report References 2

Table 3-1. SWPF Salt Batch 10, 11, 12, and 13 Density (g/mL) and Composition (mg/L (M)) (analyzed by SWPF-AL except Methyl Hg)[c]. 4

Table 3-2. SWPF Salt Batch 10 and 11 Radionuclide Activities 5

Table 3-3. SWPF Salt Batch 12 and 13 Radionuclide Activities 6

LIST OF FIGURES

Figure 4-1. Sodium and Anion Molar Concentrations for Qualification Samples (SRNL Salt Batch 5 through 9 Data and SWPF-AL Salt Batch 9 through 13 Data) 9

LIST OF ABBREVIATIONS

ARP	Actinide Removal Process
DSS	Decontaminated Salt Solution
DWPF	Defense Waste Processing Facility
IC	Ion Chromatography
IDP	Inhalation Dose Potential
ISWP	Interim Salt Waste Processing
LIMS	Laboratory Information Management System
MCU	Modular Caustic-Side Solvent Extraction Unit
MST/SS	Monosodium Titanate/Sludge Solids
SRMC	Savannah River Mission Completion
SRNL	Savannah River National Laboratory
SRS	Savannah River Site
SWPF	Salt Waste Processing Facility
SWPF-AL	Salt Waste Processing Facility-Analytical Laboratory
SPF	Saltstone Production Facility
TOC	Total Organic Carbon
VOA/SVOA	Volatile Organic Analysis/Semivolatile Organic Analysis
TTR	Technical Task Request
TTQAP	Task Technical and Quality Assurance Plan
WAC	Waste Acceptance Criteria

1.0 Introduction

This report provides a summary of the FY24 Salt Batch Qualification efforts and a comparison of those results with the previous Salt Batch Qualification results.[i, ii, iii] SRNL and SRMC analyzed four Salt Batch Waste samples from SRS, Concentration, Storage, and Transfer Facility Tank Farms Waste Tanks which included one Salt Batch Qualification sample from 41H (Salt Batch 10), one from 21H (Salt Batch 11), and two from 42 H (Salt Batches 12 and 13). SWPF-AL, for the first time, performed all short-term analyses used in the qualification of these four salt batches. SRNL performed the long-term analyses. This report details all results from SRMC and SRNL analyses performed on these qualification samples. In past years, numerous Salt Waste Batch sub-samples have been retrieved, and this data is summarized for 19 prior Salt Waste Batches (9 Salt Batch and 10 Macrobatch Qualification samples) in the previous FY21 FY22, and FY23 Summary documents.[i,ii,iii] Salt Batches were processed at SWPF, whereas the Macrobatches were processed through the Interim Salt Waste Processing (ISWP) system which included the Actinide Removal Process (ARP) and the Modular Caustic-Side Solvent Extraction Unit (MCU).

As described in the waste acceptance criteria (WAC) document,[iv] the SWPF Feed acceptance criteria ensure that the monosodium titanate/sludge solids (MST/SS) and strip effluent streams from SWPF will not exceed applicable Defense Waste Processing Facility (DWPF) safety and processing constraints (e.g., canister heat generation, nuclear criticality safety, corrosive species, and organic concentration). The SWPF Feed acceptance criteria also ensures that the decontaminated salt solution (DSS) stream from SWPF will not exceed applicable Tank 50H and Saltstone Production Facility (SPF) safety and processing constraints (e.g., inhalation dose potential (IDP)), nuclear criticality safety (fissile material and fissile poisoning), radiolytic hydrogen generation, and organic concentration.

A Task Technical and Quality Assurance Plan (TTQAP)[v] requests SRNL to provide annual technical reports of the compiled Salt Batch Qualification data to Savannah River Mission Completion (SRMC), the current Liquid Waste Operations Sub-contractor.

2.0 Experimental Procedure

2.1 Sample Receipt and Preparation

Samples of Salt Batch 10, 11, 12, and 13 were analyzed by both SRNL and SWPF-AL as described in Table 2-1. Typically, the first in the series of five 200-mL samples retrieved was transferred from H-Tank farm to SRNL. An additional four 200-mL samples were also retrieved either the same day or the next day. This second set of four samples was delivered to SWPF-AL. Transfer of samples from H-Tank farm to the respective lab was typically either the same day as the sample was retrieved or the next day but not longer than 2 days.

The details of batch compositions, mixing and sampling for these Salt Batches have been documented in X-ESR-H-01164 Rev 4.[vi] The chronology of sampling and analysis for the various batches reported herein did not necessarily follow the numerical sequence for the salt batches. Samples were retrieved on 10/2/2023 and 10/3/2023 from SRS Tank 42H for characterization of SWPF Salt Batch 12. Samples were retrieved on 12/11/2023 and 12/12/2023 from SRS Tank 41H for Salt Batch 10. Samples were retrieved on 3/11/2024 and 3/12/2024 from Tank 21H for Salt Batch 11. Samples were retrieved on 6/10/2024 and 6/11/2024 from Tank 42H for Salt Batch 13.

SWPF-AL performed all short-term (four-week) analyses for the Salt Batch 10, 11, 12 and 13 qualification samples. The Salt Batches 10, 11, 12 and 13 qualification samples analyzed by SWPF-AL were composite samples of four 200-mL tank samples listed in Table 2-1 which were combined into one 800-mL sample. SRNL provided additional long-term (eight-week) analyses on a 200-mL sample pulled in the same timeframe as the respective composite 800-mL sample analyzed by SWPF-AL (see Table 2-1)

Table 2-1. Salt Batch Sample Description and Report References

SWPF Batch	Qualification Tank	Sample(s) Pulled	Sample	References
10	41H	12/11/2023	HTF-41-23-99 (200 mL)	AL-231208-5[vii]
		12/12/2023	HTF-41-23-100/101/102/103 (composite 800 mL)	AFR-1008251, Rev. 0[viii]
11	21H	3/11/2024	HTF-21-24-37 (200 mL)	SRNL-STI-2024-00189, Rev. 1[ix]
		3/12/2024	HTF-21-24-38/39/40/41 (composite 800 mL)	AFR-1008614, Rev. 1[x]
12	42H	10/2/2023	HTF-42-23-82 (200 mL)	AL-231003-2, Rev. 1[xi]
		10/3/2023	HTF-42-23-83/84/84/86 (composite 800 mL)	AFR-1008017, Rev. 1 [xii]
13	42H	6/10/2024	HTF-42-24-59 (200 mL)	SRNL-STI-2024-00370, Rev. 0 [xiii]
		6/10/2024 and 6/11/2024	HTF-42-24-60/61/62/63 (composite 800 mL)	AFR-1008988, Rev. 0 [xiv]

For the analyses conducted in SRNL, the details of the analyses are as follows. For samples requiring filtration, filtration was conducted using a 0.45-µm nylon syringe filter and the filtrate was submitted for analysis. For the methyl mercury measurements, approximately 1.5-mL sub-samples (unfiltered) were diluted into 40 mL of ultrapure water acidified with 0.2 mL of ultrapure HCl prior to adding the 1.5 mL of sub-sample. Dilution was carried out in Teflon-capped, glass vials with very little headspace and the diluted samples were removed from the shielded cells and stored below 4 °C (in a refrigerator), following the recommended method for preparing samples of this type.[xv] For iodine analyses, aliquots of as-received sample were analyzed directly, and the separations were traced with an elemental iodine carrier from the initial aliquoting step. Samples were analyzed in duplicate or triplicate (re-test Batch 11) with a blank. The blank values were below the detection limit. The detection limit was less than 0.5% of the measured values in the samples.

2.2 Quality Assurance

Requirements for performing reviews of technical reports and the extent of review are established in Manual E7 Procedure 2.60. SRNL documents the extent and type of review using the SRNL Technical Report Design Checklist contained in WSRC-IM-2002-00011, Rev. 2. A Functional Classification of Safety Class is applied to SRNL portions of this work. A Design Verification technical review was performed via a document review according to the applicable elements in Section 5.3.1 "Design Verification by Document Review" of Manual E7 Procedure 2.60. SRNL data collection and analysis methods used in this work comply with this requirement as detailed in the TTQAP.[v] Results from this report are not governed by RW-0333P as per the TTR.[xvi]

3.0 Characterization Results for the FY24 SWPF Salt Batches (10, 11, 12 and 13) including analyses by SRNL and SWPF-AL

SWPF-AL measured and reported density of unfiltered samples (Table 3-1). SWPF-AL analyzed filtered samples for major ions including hydroxide, carbonate, formate, nitrate, nitrite, sulfate, phosphate, oxalate, chloride, and fluoride as well as elemental aluminum, sodium, potassium, and silicon (Table 3-1). SWPF-AL also reported TOC on filtered samples and total mercury analyses on unfiltered samples (Table 3-1). SRNL performed methyl mercury analyses which were performed on unfiltered samples (Table 3-1).

Radioisotopes analyzed with results determined to be above the instrument detection limit are listed in Table 3-2 and Table 3-3. Short-term analyses (Cs-137, gamma , beta and alpha measurements, U isotopes, Pu isotopes except for Pu-241) were performed by SWPF-AL, and the remaining analyses (long-term) were performed by SRNL on filtered samples. A number of additional radionuclide analyses were carried out as specified in the TTQAP.[v] Those radionuclides below the limit of detection for Salt Batches 10, 11, 12, and 13 were not included in this summary report. Analytes measured but not included in Table 3-2 and Table 3-3 are Cm-245, Ce-144, Co-60, Eu-154, Eu-155, and Sb-125. Detection limits for all analyses were below the Waste Acceptance Criteria (WAC) limits.

SRNL utilized automated Laboratoy Information Management System (LIMS) data reporting for the initial reporting of the long-term analyses data for Salt Batches 10 and 12.[vi,viii] A re-test was requested of SRNL for Salt Batch 11 to remeasure I-129 in filtered samples starting with new aliquots of original sample. This re-test was reported in a separate document.[vii] The re-test data reported herein is an average of three values measured.

Both SWPF-AL and SRNL data were used in the evaluation included in the Acceptance Report for Salt Batches 10, 11, 12, and 13.

Table 3-1. SWPF Salt Batch 10, 11, 12, and 13 Density (g/mL) and Composition (mg/L (M)) (analyzed by SWPF-AL except Methyl Hg)[c].

SWPF Batch	10[a]	11[a]	12[a]	13[a]
Density	**g/mL**	**g/mL**	**g/mL**	**g/mL**
	1.290[b]	1.299[b]	1.313	1.325
Insoluble solids	**mg/L**	**mg/L**	**mg/L**	**mg/L**
	<200	550	<200	<200
Component	**mg/L (M)**	**mg/L (M)**	**mg/L (M)**	**mg/L (M)**
Na^+	1.57E+05 (6.82)	1.67E+05 (7.27)	1.90E+05 (8.26)	1.79E+05 (7.77)
Al	7.40E+03 (0.274)	9.40E+03 (0.348)	9.60E+03 (0.356)	8.50E+03 (0.315)
K^+	1.20E+03 (0.0307)	1.42E+03 (0.0363)	1.63E+03 (0.0417)	1.34E+03 (0.0343)
Si	<5.00E+01(<0.00178)	<5.00 E+01 (<0.00178)	<5.00 E+01 (<0.00178)	<5.00 E+01 (<0.00178)
OH^-	5.30E+04 (3.12)	6.49E+04 (3.82)	6.54E+04 (3.85)	5.63E+04 (3.31)
NO_3^-	1.43E+05 (2.31)	1.08E+05 (1.74)	1.32E+05 (2.13)	1.56E+05 (2.52)
CO_3^{2-}	1.70E+04 (0.283)	1.03E+04 (0.172)	1.59E+04 (0.265)	1.44E+04 (0.240)
NO_2^-	3.56E+04 (0.774)	4.40E+04 (0.956)	5.00E+04 (1.09)	4.11E+04 (0.893)
SO_4^{2-}	4.91E+03 (0.0511)	2.24E+03 (0.0233)	3.04E+03 (0.0316)	6.39E+03 (0.0665)
PO_4^{3-}	6.30E+02 (0.00663)	3.20E+02 (0.00337)	6.40E+02 (0.00674)	4.03E+02 (0.00424)
$C_2O_4^{2-}$	1.42E+02 (0.00161)	9.40E+01 (0.00107)	2.94E+02 (0.00334)	2.05E+02 (0.00233)
HCO_2^-	<1.00E+01 (<0.00022)	<1.00E+01 (<0.00022)	8.30+01 (0.00094)	1.45E+01 (0.00016)
F^-	1.28E+01 (0.00067)	1.51E+01 (0.00079)	1.17E+02 (0.00616)	2.46 E+01 (0.00129)
Cl^-	1.98E+01 (0.00558)	2.24E+02 (0.00632)	2.32E+02 (0.00654)	2.26E+02 (0.00637)
Total Hg	5.00E+01 (0.000249)[b]	5.00E+01 (0.000249)[b]	4.50E+01 (0.000224)[b]	3.80E+01 (0.000189)[b]
Methyl Hg[c]	4.53E+00[b, c]	1.83E+01[b, c]	7.76 E+00[b, c]	2.57 E+01[b, c]
TOC	1.20E+02	9.0E+01	1.90E+02	1.40E+02

Note: molar values (M) calculated from reported values (mg/L) except for sodium in which case molar concentration (M) was the reported value.
[a] filtered unless otherwise indicated
[b] unfiltered
[c] SRNLperformed methyl mercury analyses.

Table 3-2. SWPF Salt Batch 10 and 11 Radionuclide Activities

SWPF Batch	10(filtered)	10(unfiltered)	10(filtered)	11(filtered)	11(unfiltered)	11(filtered)
Analyte	**SWPF-AL**	**SWPF-AL**	**SRNL**	**SWPF-AL**	**SWPF-AL**	**SRNL**
Units	**pCi/mL**	**pCi/mL**	**pCi/mL**	**pCi/mL**	**pCi/mL**	**pCi/mL**
^{137}Cs	3.65E+08	---	---	5.73E+08	---	---
Total gamma	---	3.02E+08	---	---	4.75E+08	---
Total beta	---	4.56E+08	---	---	7.26E+08	---
Total beta, without Cs	1.00E+06	1.09E+06	---	6.28E+05	4.35E+06	---
Total alpha	5.50E+04	6.20E+04	---	9.30E+03	6.20E+04	---
^{238}Pu	---	8.70E+04	---	---	7.50E+04	---
^{239}Pu	1.30E+03	1.29E+03	---	1.67E+02	3.30E+03	---
^{240}Pu	7.50E+02	1.02E+03	---	9.6E+01	7.40E+02	---
^{241}Pu	---	---	1.02E+04	---	---	1.23E+03
^{242}Pu	---	1.60E+00	---	---	1.19E+00	---
^{244}Pu	---	<1.00E-01	---	---	<1.00E-01	---
^{233}U	2.05E+01	2.5E+01	---	1.00E+01	1.44E+01	---
^{234}U	1.08E+02	1.15E+02	---	3.9E+01	5.0E+01	---
^{235}U	2.8E-01	3.4E-01	---	1.31E-01	2.39E-01	---
^{236}U	1.23E+00	1.49E+00	---	8.3E-01	1.00E+00	---
^{238}U	1.24E+00	1.22E+00	---	1.53E+00	2.7E+00	---
^{90}Sr	4.41E+05	4.77E+05	---	1.98E+05	1.97E+06	---
^{232}Th	---	---	<2.20E-04	---	---	5.46E+04
^{3}H	---	---	1.97E+03	---	---	7.04E+03
^{14}C	---	---	4.79E+02	---	---	1.77E+02
^{99}Tc	---	---	9.38E+04	---	---	1.15E+05
^{126}Sn	---	---	1.51E+03	---	---	1.98E+03
^{129}I	---	---	9.45E+01	---	---	1.07E+02
^{129}I Re-test[a]	---	---	---	---	---	1.55E+02
^{237}Np	---	---	7.36E+00	---	---	4.97E+00
^{241}Am	---	---	<1.11E+01	---	---	5.46E+01
^{244}Cm	---	---	2.51E+00[b]	---	---	4.35E+02

[a] re-test data reported is an average of 3 replicates
[b] replicates include values that were below detection
--- not requested

Table 3-3. SWPF Salt Batch 12 and 13 Radionuclide Activities

SWPF Batch	12(filtered)	12(unfiltered)	12(filtered)	13(filtered)	13(unfiltered)	13(filtered)
Analyte	SWPF-AL	SWPF-AL	SRNL	SWPF-AL	SWPF-AL	SRNL
Units	pCi/mL	pCi/mL	pCi/mL	pCi/mL	pCi/mL	pCi/mL
^{137}Cs	5.63E+08	---	---	4.51E+08	---	---
Total gamma	---	4.72E+08	---	---	3.79E+08	---
Total beta	---	6.72E+08	---	---	5.46E+08	---
Total beta, without Cs	1.22E+06	1.96E+06	---	8.18E+05	8.20E+05	---
Total alpha	8.60E+03	2.63E+04	---	8.00E+03	1.98E+04	---
^{238}Pu	---	2.05E+04	---	---	1.67E+04	---
^{239}Pu	2.10E+02	7.30E+02	---	3.40E+02	3.30E+02	---
^{240}Pu	1.25E+02	5.20E+02	---	1.40E+02	1.73E+02	---
^{241}Pu	---	---	5.42E+03	---	---	6.03E+03
^{242}Pu	---	5.5E-01	---	---	1.36E-01	
^{244}Pu	---	1.53E-01	---	---	<1.00E-01	
^{233}U	7.7E+00	8.2E+00	---	1.96E+01	2.09E+01	
^{234}U	5.4E+01	5.7E+01	---	5.1E+01	5.1E+01	
^{235}U	5E-01	4E-01	---	1.87E-01	1.65E-01	
^{236}U	1.50E+00	1.74E+00	---	1.03E+00	1.03E+00	
^{238}U	2.5E+00	3.2E+00	---	1.11E+00	1.24E+00	
^{90}Sr	3.28E+05	6.71E+05	---	4.02E+05	4.11E+05	
^{232}Th	---	---	4.55E-04	---	---	<5.49E-04
^{3}H	---	---	1.58E+04	---	---	5.49E+03
^{14}C	---	---	<2.70E+03	---	---	3.12E+02
^{99}Tc	---	---	1.52E+05	---	---	1.13E+05
^{126}Sn	---	---	1.81E+03	---	---	1.66E+03
^{129}I	---	---	<4.55E+01	---	---	8.48E+01
^{129}I Re-test	---	---	---	---	---	---
^{237}Np	---	---	5.06E+00	---	---	<7.07E+00
^{241}Am	---	---	2.22E+01	---	---	<5.22E+00
^{244}Cm	---	---	3.82E+01	---	---	1.95E+00

[a] re-test data reported is an average of 3 replicates
--- not requested
* replicates include values that were below detection

4.0 Comparison with Previous Salt Batch Characterization Results

SWPF Salt Batch qualification samples 10, 11, 12 and 13 were similar in chemical composition to each other and to the previous 19 salt batches analyzed to date with results trending similar to or on the higher end of the ranges previously observed for most analytes. Batches analyzed prior to Salt Batches 10, 11, 12, and 13 include ISWP Salt Macrobatches 1-10 and SWPF Salt Batches 1-9.[i,ii]

Results that were similar to previous Salt Batches included densities and insoluble solids. For Salt Batch 10, 11, 12, and 13, densities remained very close to 1.3 g/mL. The values measured for insoluble solids remained less than 200 mg/L for all batches with the exception of Batch 11, which was 550 mg/L and contained visible solids which settled with time. These solids amounts were still well below the WAC for SWPF.

Salt Batch 10, 11, 12, and 13 results for sodium, aluminum, and hydroxide were as high for these specific analyses or higher than results for salt batches analyzed to date. Salt Batches 10, 11, 12, and 13 contained 6.82 to 8.26 M Na^+ (the previously reported maximum value for all batches was 7.50 M Na^+ for Batch 6), 0.274 to 0.356 M AlO_2^- (the previously reported maximum value for all batches was 0.437 M AlO_2^- for Batch 8), 3.12 to 3.85 M OH^- (the previously reported maximum value was 4.07 M OH^- for Batch 8),[ii] 1.74 to 2.52 M NO_3^- (previous range 1.30 to 3.27 M NO_3^-), 0.77 to 1.09 M NO_2^- (previous range 0.22 to 1.28 M NO_2^-), 0.172 to 0.283 M CO_3^{2-} (previous range 0.11 to 0.39 M CO_3^{2-}), and other minor chemical components. Sulfate concentrations were measured in the range 0.0233 to 0.0675 M SO_4^2. Sulfate values for Batches 11 and 12 were lower than the previous range (0.05 to 0.16 M SO_4^{2-}). See Figure 4-1 for sodium and anion molar concentrations including SRNL data for Batches 5 through 9 and SWPF-AL data for Batches 9 through 13. For Batch 9, both SRNL and SWPF-AL data are show in Figure 4-1. Data for Salt Batches 5-9 included in Figure 4-1 is found in previous Salt Batch Qualification Summary Reports.[ii,iii]

Total mercury concentrations ranged from 38.0 to 50.0 mg/L. These mercury values are in the middle of the range reported for all prior 19 Salt Batches (9.17 to 129.0 mg/L Hg).[i,ii,iii] Methyl mercury values measured were highest for Salt Batch 13 (2.57E+01 mg/L methyl mercury). Methyl mercury for Salt Batch 13 constituted a significant fraction of the total mercury measured (3.80E+01 mg/L total mercury) in comparison to the other three batches which all had higher values measured for total mercury than Batch 13 but lower methyl mercury measured values.

Cs-137 activities in filtered samples of Salt Batches 10, 11, 12, and 13 ranged from 3.65E+08 pCi/mL (8.1E+08 dpm/mL; 1.4 Ci/gal) to 5.73E+08 pCi/mL (1.3E+09 dpm/mL; 2.2 Ci/gal). The previous maximum value was for unfiltered Salt Batch 8 (6.22E+08 pCi/mL (1.4E+09 dpm/mL; 2.4 Ci/gal)). Sample filtration had little impact on the total gamma and beta activities for the samples indicating that these activities were primarily associated with soluble species. When comparing filtered versus unfiltered samples' radionuclide measurements, typically measured values for unfiltered samples were higher than or the same as (within the uncertainty of the measurement) the measured value for the same filtered sample. This observation was most prominent for Salt Batch 11, which also had measurable solids suggesting that some radionuclides may be associated with the solids in this batch. The total beta activity was similar to the Cs-137 values measured. Ba-137m activities can be calculated from the cesium data by multiplying the Cs-137 activities by 0.946 because 94.6% of the Cs-137 decays to Ba-137m by beta decay and a secular equilibrium exists between these isotopes. The total gamma reported is lower than the calculated Ba-137m values for Batches 10, 11, 12, and 13. The total gamma activities for the filtered samples for Batches 10, 11, 12, and 13 continued to trend on the higher end of the range measured to date (4.6E+07 pCi/mL to 5.66E+08 pCi/mL (1.3E+09 dpm/mL; 2.14 Ci/gal)). Revisions continue to be made to the SWPF Waste Acceptance Criteria (WAC) requirement related to Cs-137 content. Around the time of these qualification samples, the SWPF WAC for Cs-137 content had been raised to 5.1 Ci/gal but has subsequently been lowered to 3.0 Ci/gal (uncertainty of two sigma applies) based on SWPF Safety Basis calculations related to hydrogen generation where Cs-137 value is an input.[xvii] The current SWPF WAC limit for Cs-137 is 3.0 Ci/gal.[iv] The Cs-137 values for Salt Batches 10,11, 12, and 13 remained below the current WAC value of 3.0 Ci/gal.

There were a number of analyses requested as a part of the TTQAP[v] that were not measured at values above the instrument detection limit and are not included in this report. These analyses include Cm-245, Ce-144, Co-60, Eu-154, Eu-155, and Sb-125.

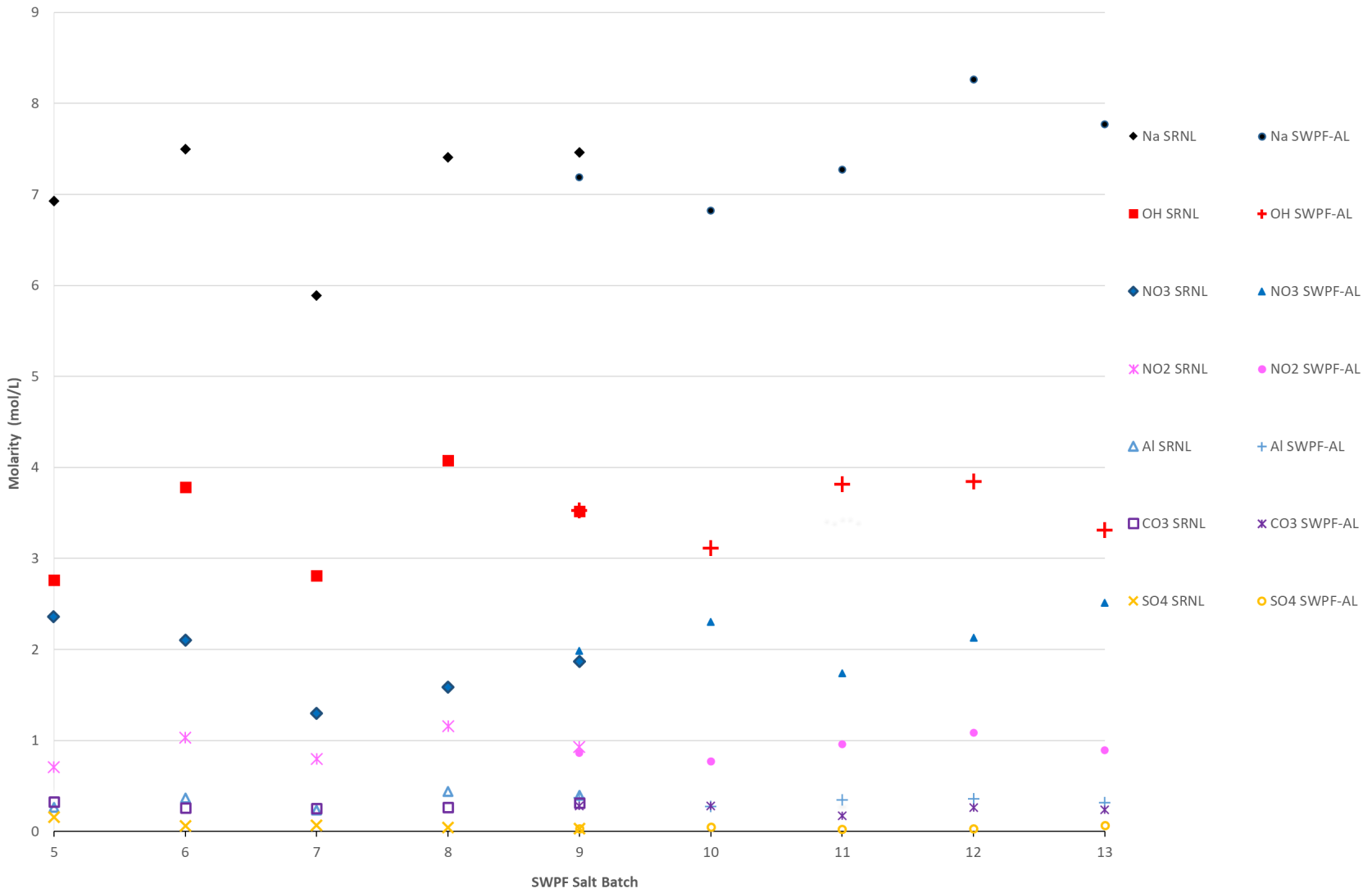


Figure 4-1. Sodium and Anion Molar Concentrations for Qualification Samples (SRNL Salt Batch 5 through 9 Data and SWPF-AL Salt Batch 9 through 13 Data)

As compared to the previous 19 Batches, the trends following cesium removal are similar for Batches 10, 11, 12, and 13. Total beta activity for the samples decreased dramatically (two orders of magnitude) as compared to Cs-137 values. Activities for radionuclides other than Cs-137 and Ba-137m in the samples were primarily associated with Sr-90/Y-90, Tc-99, Pu-238, and Pu-241. Although I-129 activities (<45.5 pCi/mL to 155 pCi/mL) are relatively low in comparison to the other radionuclides reported herein, the measured value for Batch 11 (155 pCi/mL) was the highest value measured for batches 10 through 13 and close to the previous maximum measured for Batch 8 (167 pCi/mL). I-129 and Tc-99 (9.38E+04 pCi/mL to 1.52E+05 pCi/mL) values continue to trend near the WAC limits for the Saltstone Production Facility (SPF) which are currently 1.00E+02 and 2.11E+05 pCi/mL, respectively.[xviii]

Although I-129 values did not exceed the SWPF WAC and do not impact the qualification of these Salt Batches or their processing through SWPF, additional measures are in place to reevaluate I-129 prior to transferring to Tank 49 as a part of an active deviation for Salt Batches 8-22 to justify exceeding the SPF WAC limit if it is not met at that time.

Pu is the primary alpha emitter in tank waste. The plutonium isotopes Pu-238, Pu-239, and Pu-240 are the major contributors to the alpha activity in SRS Salt Batch Feed samples, and Pu-238 accounts for the majority of the alpha activity in Salt Batch Samples 10, 11, 12, and 13. Pu-238 activity, of unfiltered samples, ranged in value from 1.67E+04 pCi/mL (3.71E+04 dpm/mL) for Batch 13 to 8.70E+04 pCi/mL (1.93E+05 dpm/mL) for Batch 10. The maximum Pu-238 values reported previously[ii] were 8.39E+04 pCi/mL (1.9E+05 dpm/mL) for filtered (Batch 5) and 9.14E+04 pCi/mL (2.0E+05 dpm/mL) for unfiltered (Batch 5). For Pu isotopes, where both unfiltered and filtered samples were measured, the unfiltered sample was higher than the filtered sample; this difference was most significant for Batch 11 where solids were observed. These activities are at or near the values for the total alpha with cesium removed method. The total alpha measurement following cesium removal is a gross semi-quantitative analysis (suffering from the high beta to alpha activity ratio in Salt Batch matrices) and has poor precision compared to the sum of the quantitative radiochemical analyses results for the various actinides.

5.0 Conclusions

None of the samples displayed any unusual or unexpected visual characteristics. In general, the FY24 Salt Batch Samples analyzed in Batches 10, 11, 12, and 13 were similar to each other in chemical and radiological composition and followed similar trends for filtered, unfiltered, and samples with cesium removal prior to analysis. Total beta measurements with cesium removed unfiltered and filtered values were similar with the exception of Batch 11 where the presence of solids could be attributed to the difference in these unfiltered and filtered values. For the Pu and U isotopes and total alpha analyses where both unfiltered and filtered values were measured, a significant difference in unfiltered versus filtered values was observed in general and was greatest for Batch 11 (either close to or over an order of magnitude for Pu-239, Pu-240, total alpha, and Sr-90 measurements). These observations indicate that some of these species in Batch 11 may be insoluble and associated with the insoluble solids which were measured to be higher in Batch 11 than in Batches 10, 12, or 13. Sample filtration had little impact on the total beta without cesium activities for the samples indicating that these activities were primarily associated with soluble species except for Batch 11 which contained higher solids. Values for a number of analytes measured in Batches 10, 11, 12, and 13 continued to trend toward values as high or higher than previously reported for the 19 previous Salt Batches (10 Macrobatch samples and 9 Salt Batch Samples). These analyses trending higher included sodium, aluminum, and hydroxide as well as Cs-137 (and related analyses for total gamma). Although the concentrations and activities were higher in value and exceeded targets included in the SWPF WAC for sodium, aluminum, and hydroxide and the SPF WAC limit for I-129, no significant impact was deduced from the results based on evaluations carried out to justify any WAC deviations.

6.0 Recommendations, Path Forward, or Future Work

In the next summary document, the average values and ranges for key measured analytes for all Salt Batches reported to date through Salt Batch 13 (10 Macrobatch plus 13 Salt Batch samples) should be updated in order to be compared with the FY25 Salt Batch sample data.

Sulfate values for several batches (11 and 12) were noted to be lower than the previous range observed for all salt batches. Precipitation of insoluble sulfate species (such as kogarkoite) is anticipated at higher sodium concentration. OLI modeling can be used to further explore the behavior of various insoluble species as a function of changing sodium or other ion concentrations to predict the behavior of insoluble species in salt batch heels.

Similar to this annual summary report for the Salt Batch Qualification feed, a periodic summary document to consolidate and summarize DSS data would be useful as a consolidated reference for comparison of the DSS data with the salt batch data in the Annual Salt Batch summary documents.

7.0 References

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[iv] X-ESR-J-00001, Rev. 10, Salt Waste Processing Facility Feed Waste Acceptance Criteria, Savannah River Mission Completion, Aiken, South Carolina. July 11, 2024.

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[vii] Peters, T. B., External SWPF Sample Results Report, AL-231208-5, February 2024.

[viii] Rudder, C., SWPF Analytical Laboratory Analytical Final Report, AFR-1008251, Rev 0, January 2024.

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[xviii] X-SD-Z-00004, Rev. 5, Waste Acceptance Criteria for Transfers to the Z-Area Saltstone Production Facility, Savannah River Mission Completion, Aiken, South Carolina. February 28, 2024

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